



Applied Microeconomics: Public economics

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Public Economics

Recall from first-year micro:

First fundamental theorem of welfare:

In perfectly competitive markets, efficiency is maximised

- All possible gains from trade exploited
- Adam Smith's 'invisible hand'

Does this imply that:

1. Markets always lead to efficient outcomes?
2. No role for the government in the economy?
3. We only care about efficiency?

Public Economics

Is it efficient that

- Large-scale fishing drove some species close to extinction?
- Rising sea levels threaten to overflow the Maldives?
- In the US, millions of people did not have health insurance?

And if markets are so efficient, then why

- Is car insurance mandatory in many countries?
- Do you not pay the full cost of your education?
- Is most infrastructure not owned by the private sector?
- Do governments impose lockdowns during a pandemic and provide vaccines for free?

Public Economics

Note:

Even under perfect competition, government is needed:

→ Enforcement of property rights!

Public Economics

When markets lead to undesirable outcomes, governments **may sometimes** be able to improve upon market outcome

Means to intervene:

- Public provision
(education; infrastructure; research; vaccines)
- Affecting prices through taxes, excises, and subsidies
(cigarettes; gasoline; solar power; soft drink sugar)
- Regulation
(Fishing; car insurance; labour; CFCs; guns; prostitution; organs; babies)
- Public production
(defense; street lights; income insurance; prisons)

Public Economics

First part of this course:

1. In which situations do markets systematically *not* yield the socially efficient outcome, and why? → **market failure**
2. Can private parties improve upon the market outcome?
3. Can government intervention improve the outcome?
4. Does government intervention yield the socially efficient outcome? → **government failure**

Second part:

1. Redistribution and taxation
→ Trade-off between efficiency and distribution
2. Collective decision-making & government failure

Organisation Public Economics

- Lecturer: Dr. Josse Delfgaauw
- Tutorials: Alicja Samełko, Bastiaan Fennema, Axel Vrolijk,
Luigi Zucchini Solimei
- Literature: - Rosen & Gayer, *Public Finance*,
8th ed. **or** 9th ed. **or** 10th ed.
- Lecture material & exercises
- Canvas: Course setup; slides; exercises; old exams
- Grading: Exam (100%)

This week

Today:

- Quick review of Perfect Competition → video
- Short introduction of market failures
- Public goods

Next lecture (Friday):

- Beyond efficiency
- Public versus private provision: the key trade-offs

Quick review of Perfect Competition

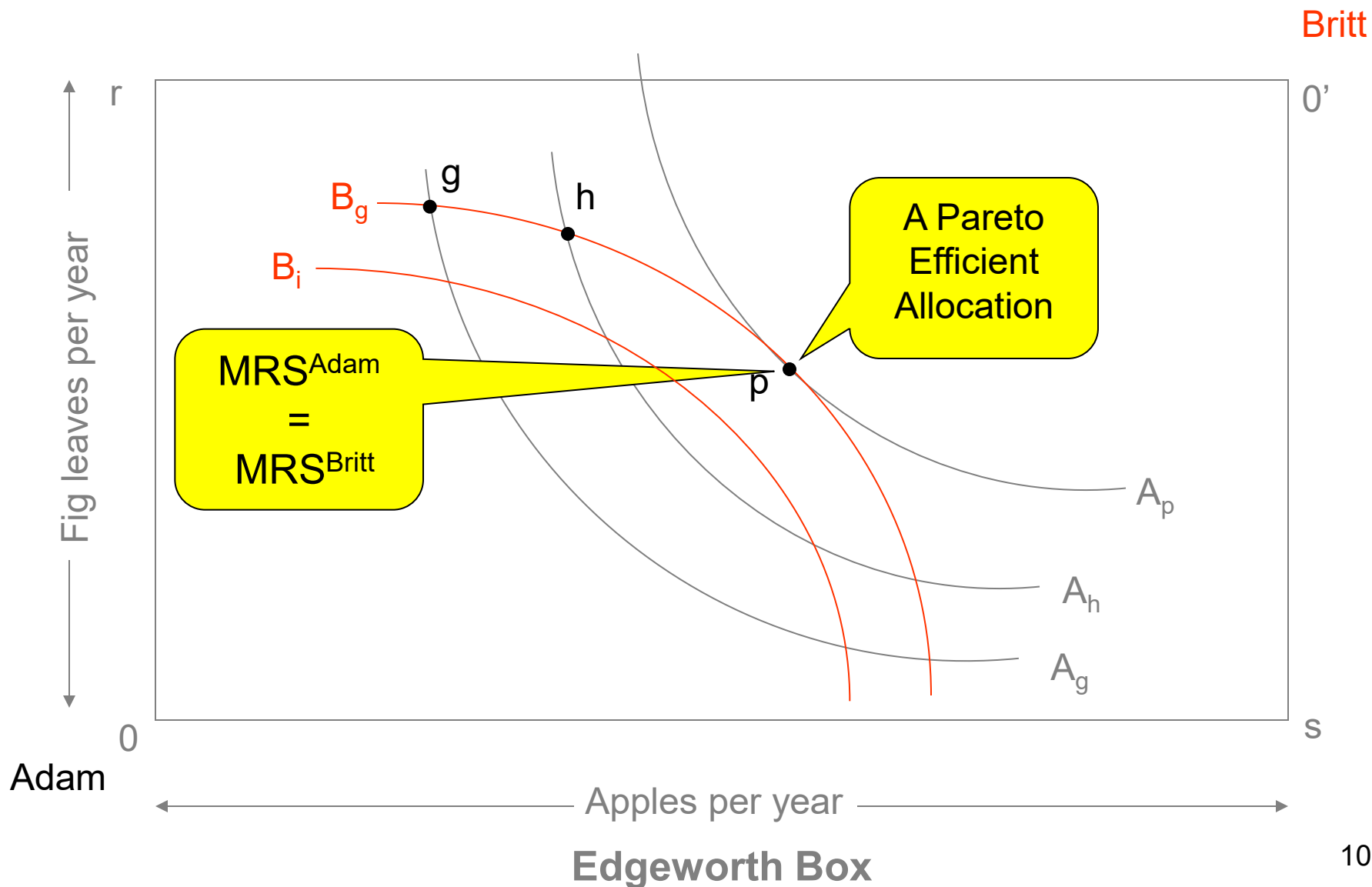
Two good - two person (A & B) exchange economy

With standard indifference curves, efficiency is maximised when $MRS^A = MRS^B$

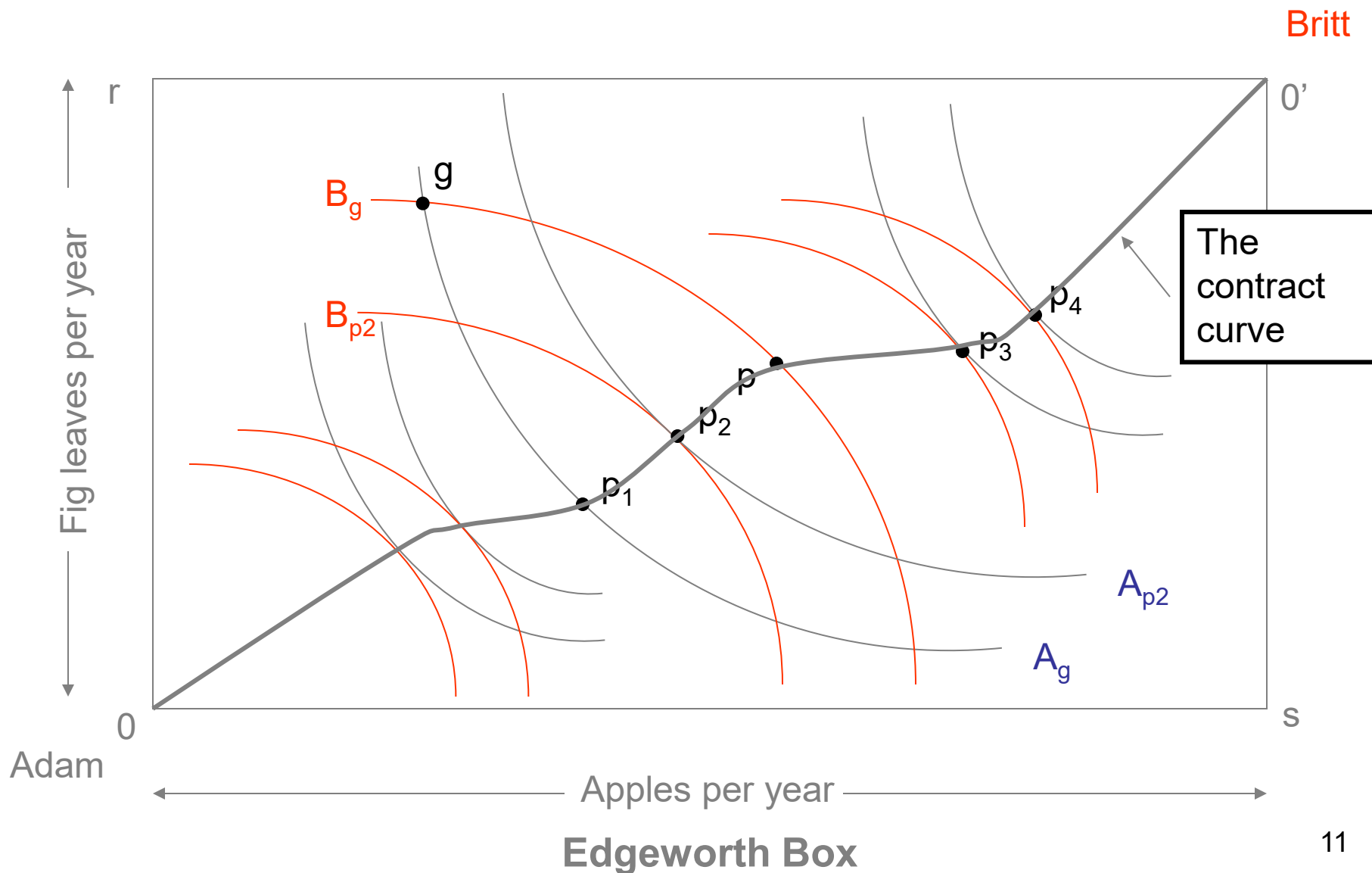
- Recall: MRS gives ratio at which trading goods does not make individual better or worse off.
→ How much additional good Y is needed to compensate for a reduction in good X by 1
- If $MRS^A \neq MRS^B$, trade can make one better off without making other worse off → Pareto-improvement

First and Second FTW

Quick review of Perfect Competition



Quick review of Perfect Competition



Quick review of Perfect Competition

Add production side: Production Possibilities Frontier

- | Slope PPF | = Marginal Rate of Transformation
- MRT: how much additional good Y can be produced when production of good X is reduced by 1
- $MRT = MC^X / MC^Y$

In pareto-efficient equilibrium:

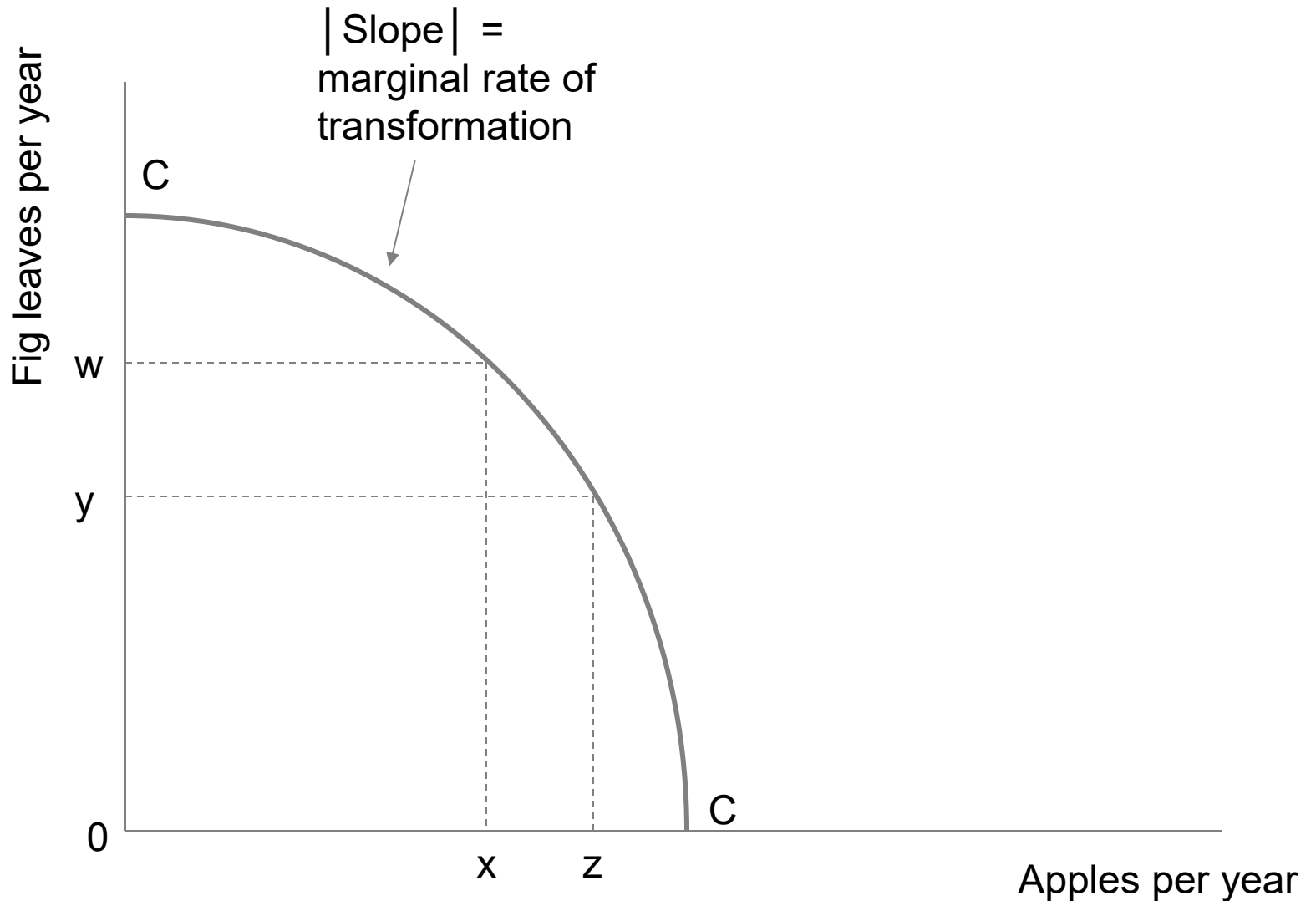
$$MRS^A = MRS^B = MRT$$

- If $MRS^A \neq MRT$, different ‘product mix’ improves efficiency

Mechanism: under PC, prices adjust to get

$$MRS^A = MRS^B = MRT = p^X / p^Y$$

Production Possibilities Curve



Market Failure

Four main reasons for market failure:

1. Market power
 2. Public goods
 3. Externalities
 4. Asymmetric information
- 
- ‘Missing markets’

First part of this course:

Why markets are not efficient in case of market failure,
and how efficiency may be improved.

Government failure

Normative appeal of government intervention can be limited by problems in collective decision-making: government failures

Four main reasons for government failure:

1. Lack of information (about individuals' preferences and production processes)
2. Imperfect political representation and problems in aggregating preferences (Arrow's impossibility theorem)
3. Rent-seeking and corruption
4. Limited or misaligned incentives

Week 7 of the course:

Positive analysis of collective decision-making

Remainder of this lecture: public goods

1. What is a public good?
2. Characteristics of (pure) public goods
 - Non-rival
 - Non-excludable
3. Efficient provision of public goods
4. Inefficiency of market (= private) provision
 - Free-rider behaviour
 - underprovision
5. Problems in public provision of public goods
 - Misrepresentation of preferences
 - overprovision

Private versus Public Goods

- Pizza
- House
- Highway with traffic jam
- Health care
- Defense
- Internet
- Highway without traffic jam
- Lighthouse

*“If I consume a unit of the good, someone else **cannot** consume **that same unit**”*

*“If I consume a unit of the good, someone else **can** consume **that same unit as well**”*

Characteristics of ‘pure’ public goods

1. Non-rival in consumption

Consumption by A does not prevent consumption by B.

→ Once it is provided, additional consumers do not raise the cost of provision

→ MC of (additional) consumer = 0.

2. Non-excludable

It is not (or hardly) possible to prevent any person from consuming the good.

Impure public goods

Illustration non-excludability: lighthouse

- Two harbors share one dangerous spot; no communication.
- Lighthouse: Construction cost: 15 (shared if both contribute).
Benefit to each: 10 (non-excludable)
- Question: In Nash equil., will the lighthouse be constructed?
- Answer: No (unless they find means to credibly commit to share construction cost → role for government)
→ Free-rider behaviour

		H2	
H1		Construct	Not construct
	Construct	(2.5, 2.5)	(-5, 10)
	Not construct	(10, -5)	(0, 0)

Illustration (Non-)Rivalness

Example: traffic jams

On most roads, usage is free: $p = 0$

Outside rush hour: road use non-rival: $MC = 0$

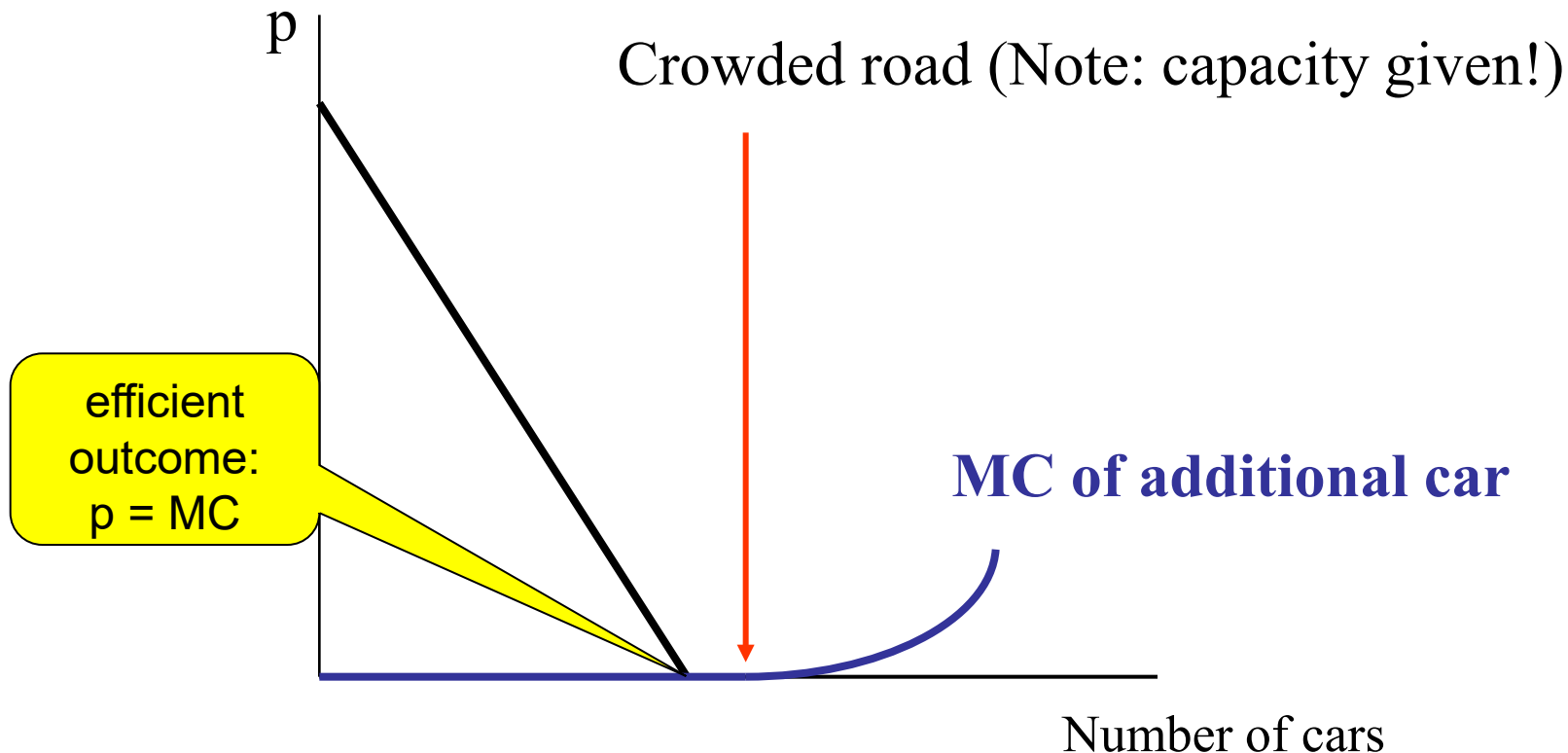
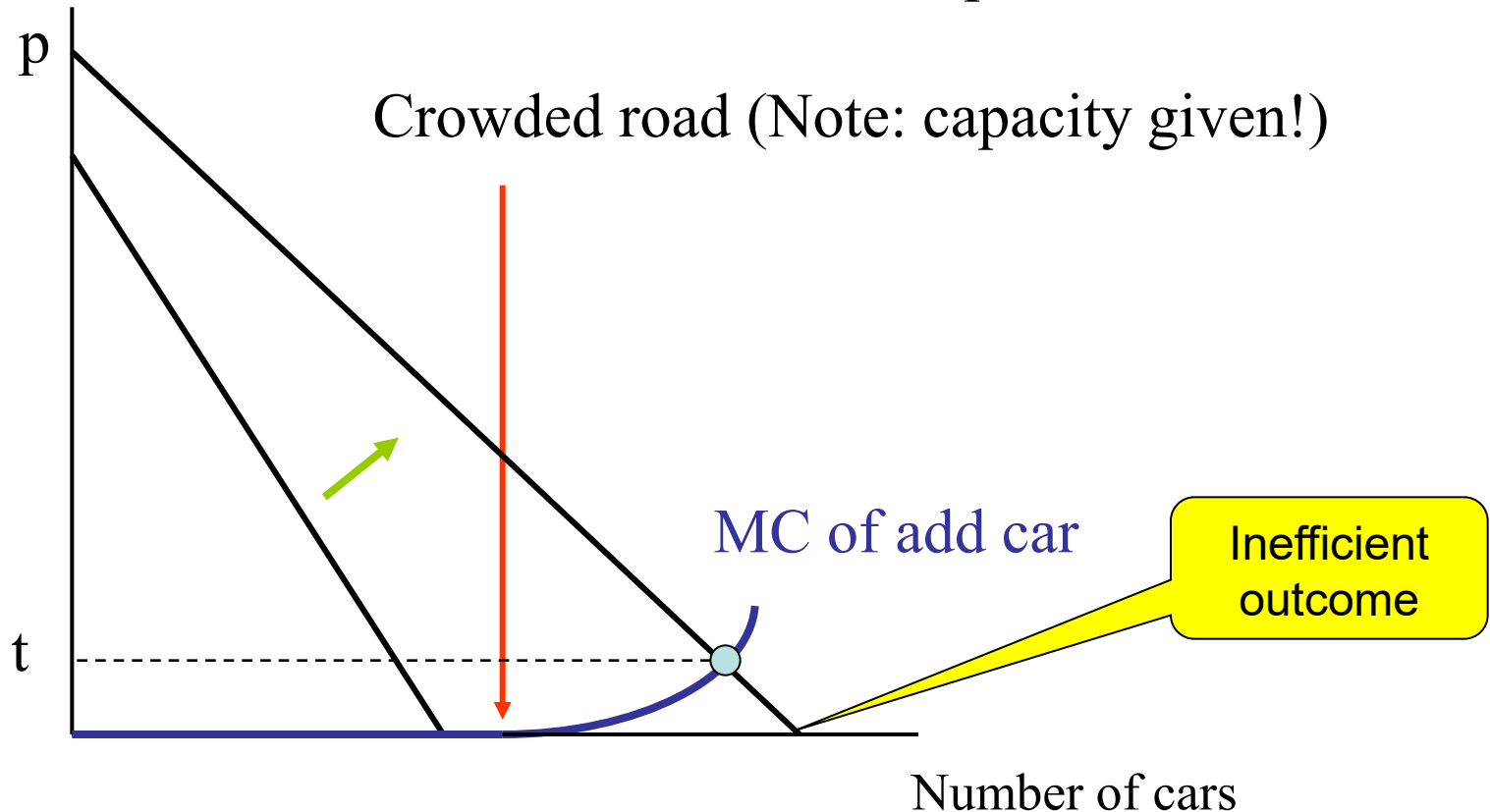


Illustration (Non-)Rivalness

Higher demand during rush hour; makes road use rival: $MC > 0$

Outcome inefficient: $p = 0 < MC$



Excludable? Yes, by flexible road-pricing: $t = p = MC$ in rush hour.
Yields socially efficient outcome (but involves transaction cost) 21

Efficient provision of public goods

Recall: With private goods, societies' marginal valuation for an additional unit equals the individuals' marginal valuation (= MRS)

In case of public goods:

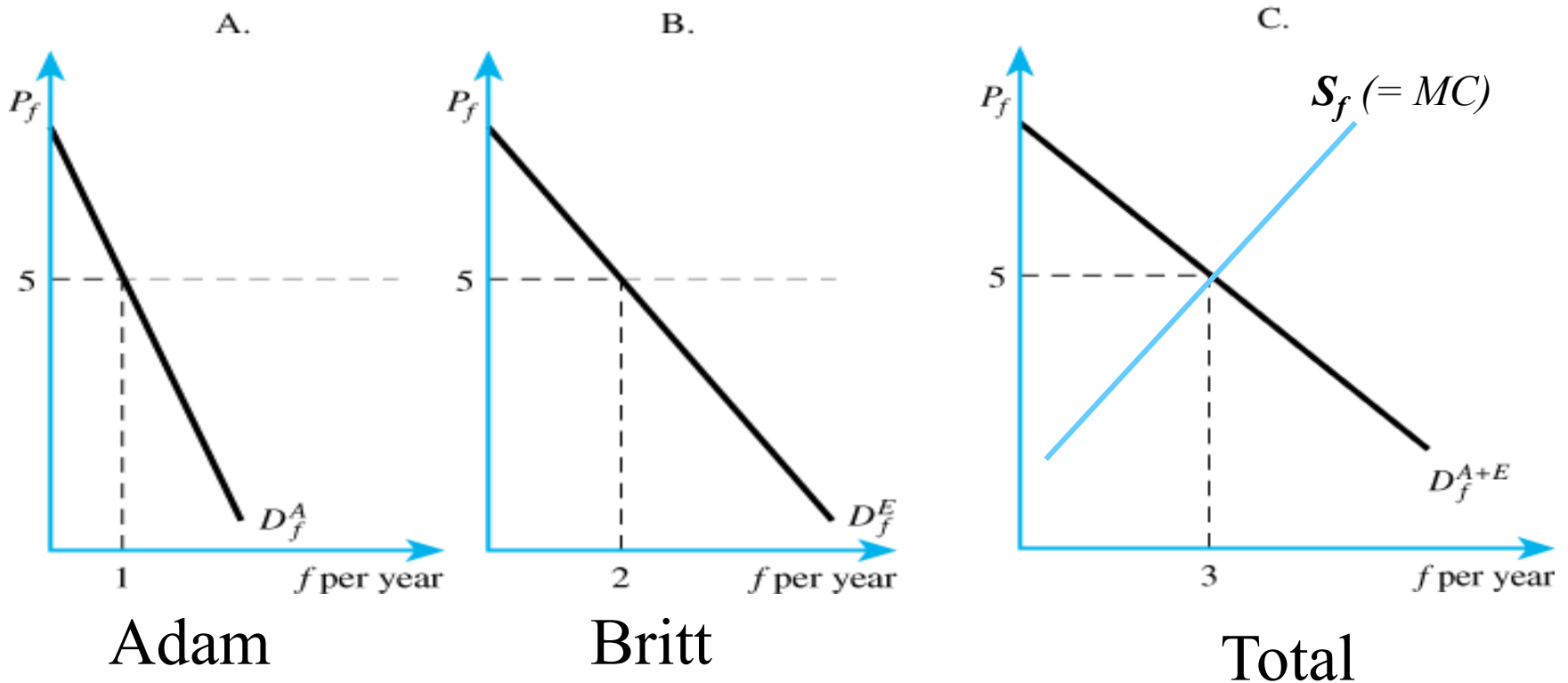
Non-rival characteristic implies everyone consumes all units of the public good

→ So, to find societies' marginal valuation for (additional) units, we must sum over individuals' marginal valuations

Notes

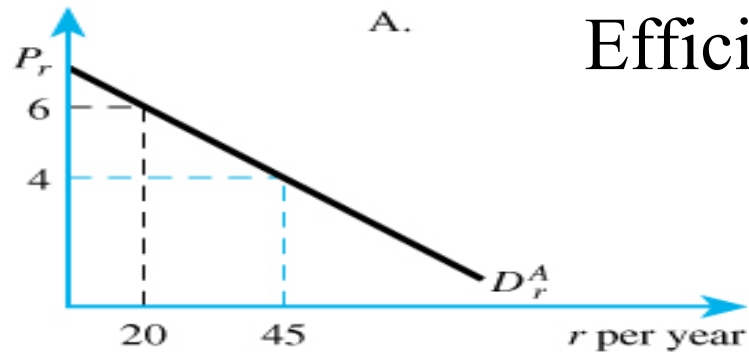
In following figures, price apples assumed equal to 1, so that MC apples also 1
Then, demand curve fig leaves / rockets gives MRS, supply curve gives MRT

Efficient provision of private good

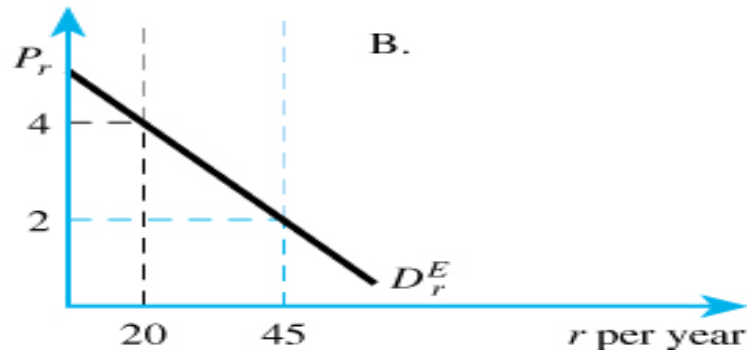


For total demand: Horizontal summation

Efficient provision of public good

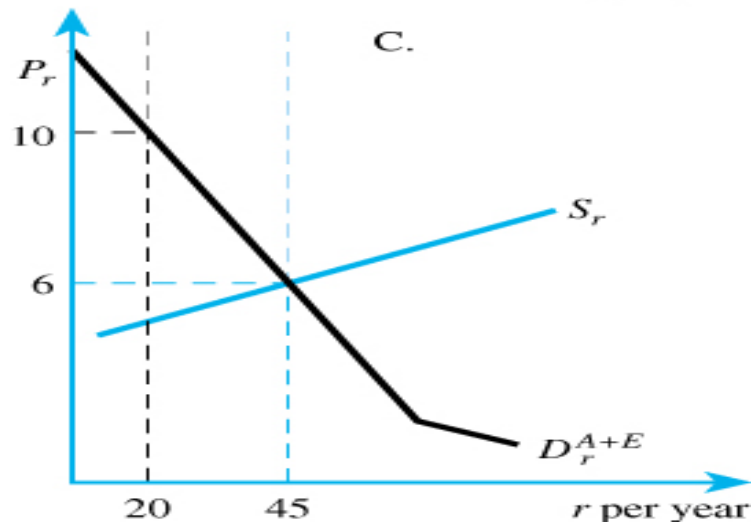


Adam



Britt

For total demand:
Vertical summation



Total

Efficient provision: private vs public goods

Recall: efficient provision of *private* good:

$$MRS^A = MRS^B = MRT$$

Efficient provision for *public* good:

$$MRS^A + MRS^B = MRT$$

- **Total** valuation of last (marginal) unit must be equal to the social cost of providing this last unit
→ ('Samuelson condition')

Efficient provision: private vs public goods

Private goods

- Supply equals total demand
- Quantity consumed differs among people
- Everyone pays the same price, and has the same marginal valuation (= MRS)
- Market generates Pareto-efficient equilibrium

Public goods

- Supply equals total demand
- Everyone consumes the same quantity
- People have different marginal valuations
- Market does not provide efficient outcome
→ **Market failure**

Problems in private provision of public goods

Non-excludability implies:

- People benefit from public goods even without contributing
→ no-one can be 'forced' to pay for consumption of public good provided by others

So, everyone prefers that others pay for public goods

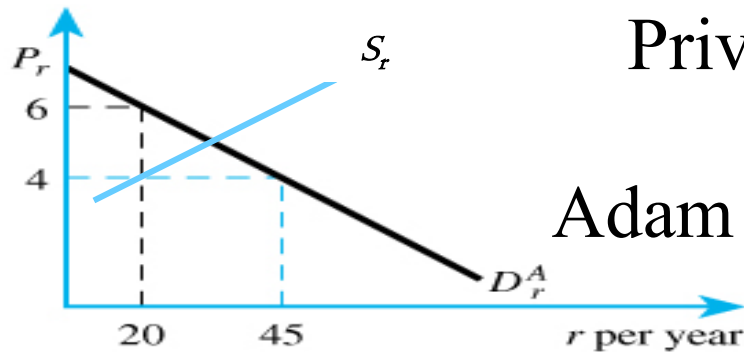
Non-rivalness implies:

- If someone contributes, other people benefit
- But: when contributing, people may not (fully) take into account that others also benefit from their contribution

Result: Free rider behaviour, leading to underprovision.

→ Market failure

Private provision of public good



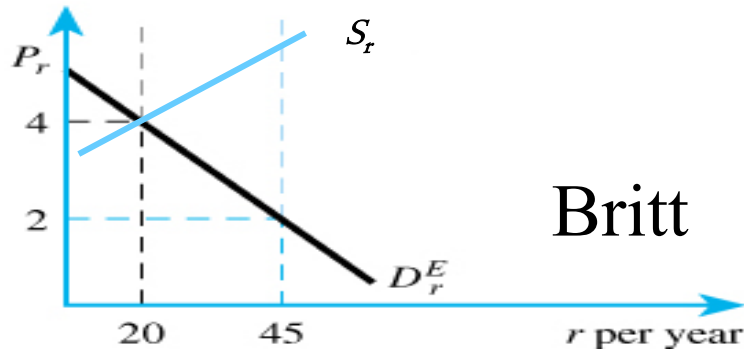
Efficient: $r = 45$
costs $6 \cdot 45 = 270$

Suppose:

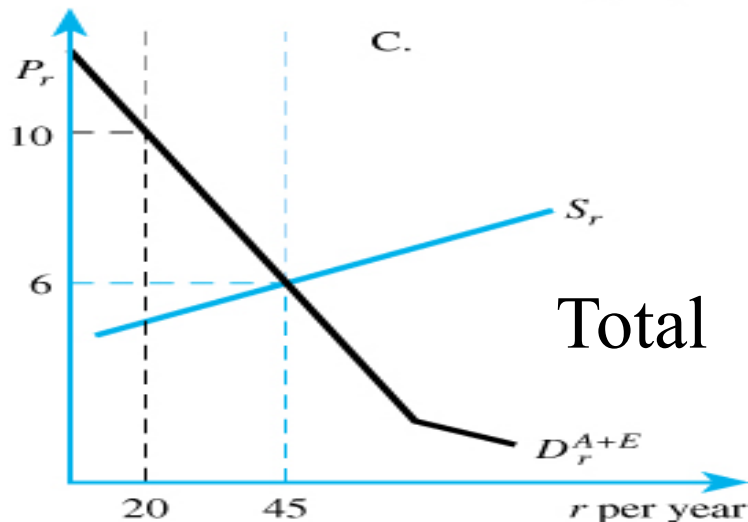
Adam pays $4 \cdot 45 = 180$

Britt pays $2 \cdot 45 = 90$

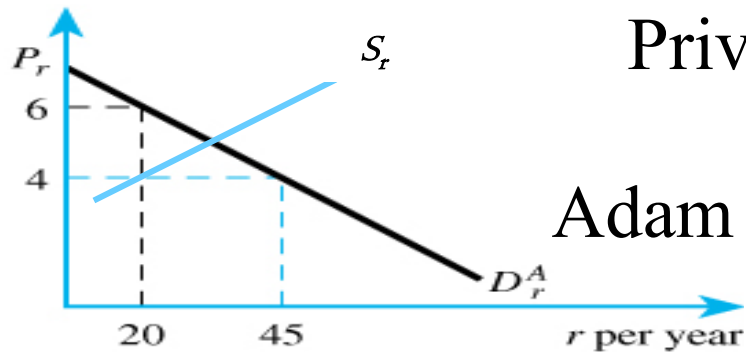
But then Britt realises that Adam effectively pays for $180/6 = 30$ rockets



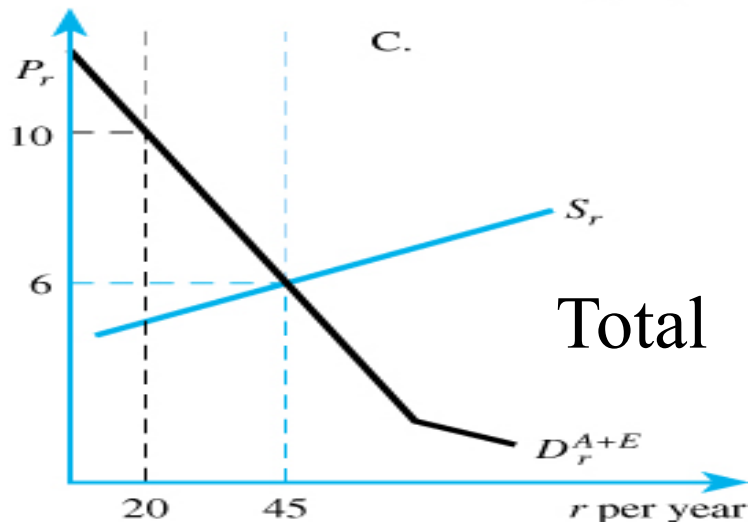
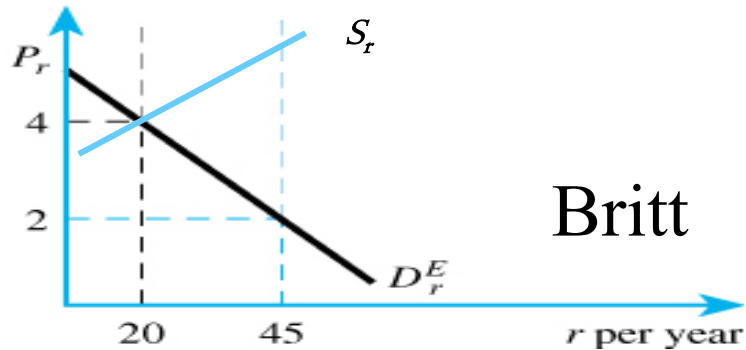
→ Britt better off paying zero:
marg cost of rocket at $r = 30$ larger than Britt's marg benefit



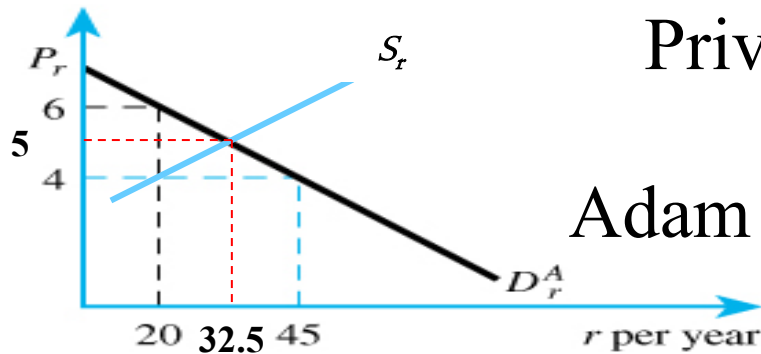
Private provision of public good



Note: Same goes for Adam:
 If Britt contributes 90, Adam is not willing to pay 180:
 his marg benefit of 45e rocket (=4)
 is below marg cost (=6)



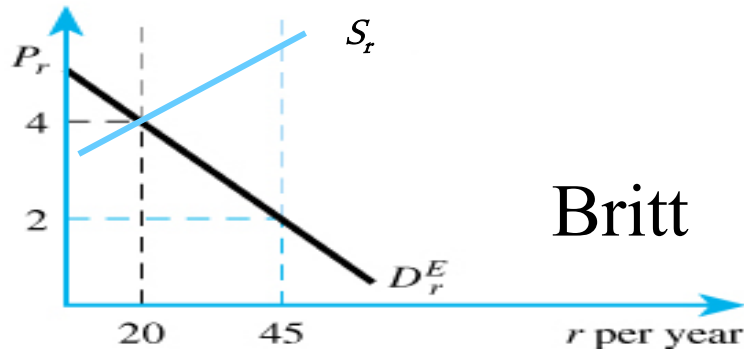
Private provision of public good



Only Nash Equilibrium here:

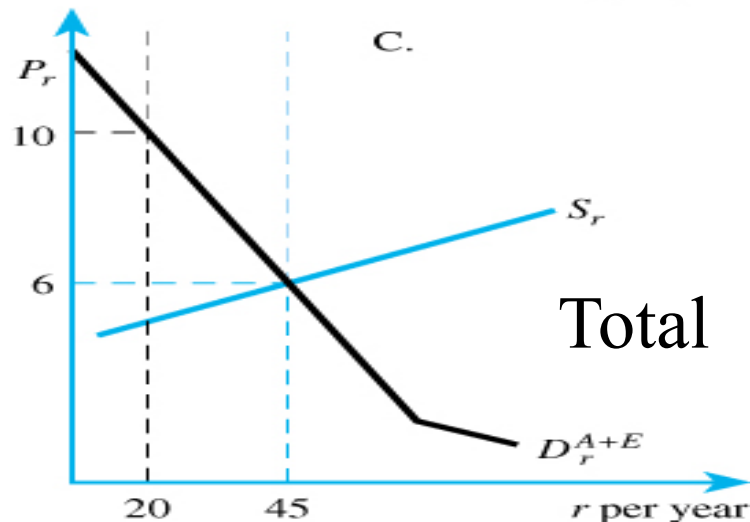
- Britt contributes nothing
- Adam buys the number of rockets that maximises his own utility,
→ Here 32.5 (at price 5)

→ Underprovision: Inefficient



Note that Adam contributes 0 and Britt buys 20 rockets is **NOT** a Nash equilibrium

→ If Britt buys $r = 20$, Adam prefers to buy 12.5 more rockets over buying 0.



Problems in private provision of public goods

Efficiency loss due to underprovision larger in larger groups

Example:

N identical individuals, who can all contribute max 1 unit of a public good.

Let G be the total number of units of the public good provided

Marginal cost per unit is constant and equals $p > 0$

Utility of individual i equals $U_i(G) = V \ln(G) - p g_i$,
where V is a parameter and $g_i \in \{0,1\}$.

$$\rightarrow MU_i = \frac{V}{G}$$

Efficient level of G ? $\sum_i MU_i = p$

$$N \frac{V}{G} = p \rightarrow G = \frac{NV}{p}$$

Problems in private provision of public goods

Now suppose all individuals decide independently whether to buy one unit of the public good

What is the Nash equilibrium outcome on this market?

For any individual it is optimal to buy one unit if and only if

$$MU_i \geq p \rightarrow \frac{V}{G} \geq p$$

Hence, the Nash equilibrium outcome has $G = \frac{V}{p}$

→ Market failure as soon as $N > 1$

→ Efficiency loss increasing in N

(math derivation on next slide not part of exam material)

Problems in private provision of public goods

Compare **socially efficient** and **private (Nash)** provision

$$G = \frac{NV}{p} \text{ versus } G = \frac{V}{p}$$

Efficiency loss due to market failure equals

$$\sum_i U_i \left(\frac{NV}{p} \right) - \sum_i U_i \left(\frac{V}{p} \right) \\ NV \left(\ln \frac{NV}{p} \right) - p \frac{NV}{p} - \left[NV \left(\ln \frac{V}{p} \right) - p \frac{V}{p} \right]$$

This can be rewritten as

$$NV \left(\ln \frac{NV}{p} - \ln \frac{V}{p} \right) - V(N - 1) = NV(\ln N - 1) + V$$

Which increases in N !

→ Generally, efficiency loss larger in larger populations

Problems in private provision of public goods

Reality check: In (repeated) public good experiments,

- Some people do ‘donate’ significant amounts
- Donations decrease in later sessions
- Punishment option increases donations
- (People punish even when punishing is costly)

(Based on Fehr and Gächter, 2000, AER)

Problems in private provision of public goods

? Solution to free-rider behaviour: make consumption excludable and charge ‘admission fee’ ?

But:

1. Making good excludable may be costly
2. Non-rivalness implies that also people with very low valuations should consume → fee should be zero

? Alternative: personalised prices ?

- High valuation – high p ; low valuation – low p
- Perfect price discrimination; Lindahl prices (see R&G)
- But then: incentive to lie about valuation
- Back at free-rider problem!

Problems in public provision of public goods

Is provision by the government easy solution?

No!

1. Government does not know individuals' preferences.
 - If 'society' pays, individuals have incentive to exaggerate their (marginal) valuation of the public goods
 - Overprovision!
 - (theoretical solution: preference-revelation mechanisms)
2. Government must raise taxes to finance public goods provision
 - Lowers efficiency through tax distortions (week 5-6)

Looking ahead

Friday:

- Extensive review of Perfect Competition → video
- Beyond Efficiency
- Key issues in trade-off public vs private provision

Next week:

- Externalities